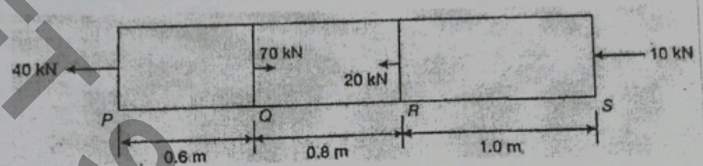
DEVI AHILYA UNIVERSITY, INDORE
II BE (Civil Engineering)
Class Test II, Octomber 2023
Subject Code : 3VLRC2
Subject Name : Applied Mechanics and Strength of materials

Duration: 70 Min | Maximum Marks: 20

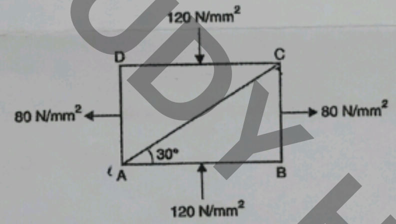
Note: Attempt any four questions. All questions carry equal marks. Assume Any Data Missing.

Q1. A tapering rod has diameter d1 at one end and it tapers uniformly to a diameter d2 at the other end in a length L. If modulus of elasticity of the material is E. Derive an expression for its change in length when subjected to an axial force P.

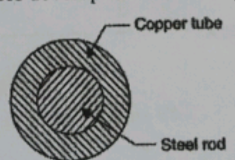
Q2. A brass bar, having a cross sectional area of 900 mm^2 is subjected to axial forces as shown in fig below. find total elongation of bar. Take $E = 1 \times 10^5$ N/mm^2



Q 3. The direct stresses at a point in the strained material are 120 N/mm2 compressive and 80 N/mm2 tensile as shown in Fig. There is no shear stress. Find the normal and tangential stresses on the plane AC. Also find the resultant stress on AC.



Q4. A compound bar consists of a circular rod of steel of 25 mm diameter rigidly fixed into a copper tube of internal diameter 25 mm and external diameter 40 mm as shown in Fig. If the compound bar is subjected to a load of 120 kN, find the stresses developed in the two materials.



Q5. Derive an expression for Normal and Tangential stresses on an inclined plane subjected to two mutually perpendicular direct stresses.